

1.1 Atomic Structure

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.1 Atomic Structure
Difficulty	Medium

Time allowed: 70

Score: /53

Percentage: /100

Question 1a

The composition of atoms and ions can be determined from knowledge of atomic number, nucleon number and charge.

Complete Table 1.1.

Table 1.1

Atomic number	Nucleon number	Number of electrons	Number of protons	Number of neutrons	Symbol
9		10			${}^{19}_{9}\text{F}^{-}$
		23	26	32	

[2 marks]

Question 1b

Lithium and potassium are Group 1 metals.

State the following for the potassium 1+ ion with a nucleon number of 39.

Symbol

[1]

Full electronic configuration

[1]

[2 marks]

Question 1c

Beams of protons, neutrons and electrons behave differently in an electric field due to their differing properties.

Fig. 1.1 shows the path of a beam of electrons in an electric field.

Add and label lines to Fig. 1.1 to represent the paths of beams of protons and neutrons in the same field.

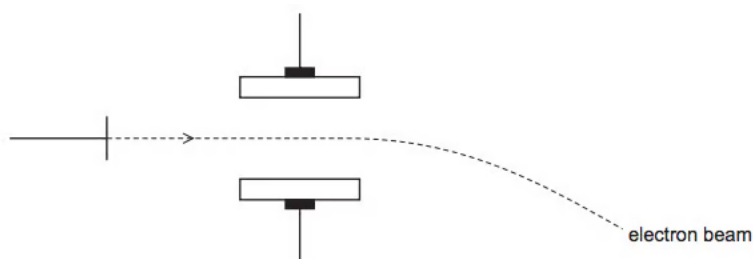


Fig. 1.1

[3 marks]

Question 1d

The fifth to eighth ionisation energies of three elements in the third period of the Periodic Table are given. The symbols used for reference are not the actual symbols of the elements.

	Ionisation energies, kJ mol^{-1}			
	fifth	sixth	seventh	eighth
X	6274	21 269	25 398	29 855
Y	7012	8496	27 107	31 671
Z	6542	9362	11 018	33 606

i)

State and explain the group number of element **Y**.

Group number

Explanation

[1]

ii)

State and explain the general trend in first ionisation energies across the third period.

[2]

iii)

Explain why the first ionisation energy of element **Y** is less than that of element **X**.

[2]

iv)

Complete the electronic configuration of element **Z**.

$1s^2$

[1]

[6 marks]

Question 2a

The model of the nuclear atom was first proposed by Ernest Rutherford. He developed this model on the basis of results obtained from an experiment using gold metal foil.

Complete the Table 2.1 with information for two of the particles in an atom of ^{197}Au .

Table 2.1

Particle	Relative mass	Relative charge	Location within atom	Total number in an atom of ^{197}Au
electron	0.0005	-1		79
neutron			nucleus	

[4 marks]

Question 2b

A sample of gold found in the earth consists of only one isotope.

i)

Explain what is meant by the term isotopes

[2]

ii)

A different sample of gold contains more than one isotope.

Suggest why this different sample of gold has the same chemical properties as the sample found in the earth.

[1]

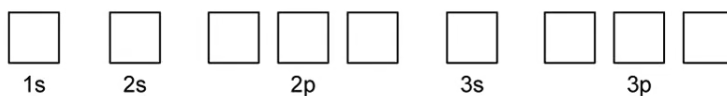
[3 marks]

Question 2c

Gold(III) chloride, traditionally called auric chloride, is one of the most common compounds of gold.

It has the formula AuCl_3 .

Complete the electron configuration for the chloride ion using box notation.



[1 mark]

Question 2d

On the axes shown in Fig. 2.1, draw a sketch diagram of one of each different type of orbital that is occupied by the electrons in a Period 3 element.

Label each type of orbital.

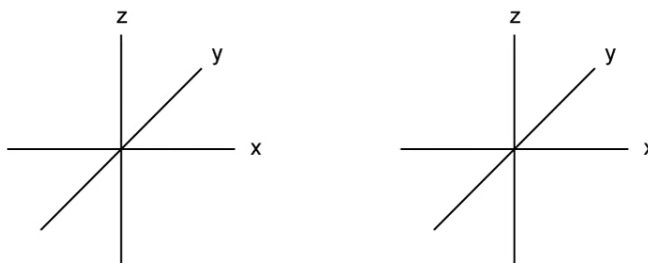


Fig. 2.1

[3 marks]

Question 3a

All elements have a value for a first ionisation energy.

i)

Define the term first ionisation energy of an element.

[2]

ii)

Write the equation for the first ionisation energy of aluminium.

[1]

[3 marks]

Question 3b

Table 3.1 shows successive ionisation energies of an element **A**, found in period 3 of the Periodic Table.

Table 3.1

Number of electrons	1	2	3	4	5	6	7	8
Ionisation energy (kJ mol^{-1})	1012	1904	2914	4964	6274	21 268	25 431	29 872

Identify element **A**. Explain your answer using data from Table 3.1.

[2 marks]

Question 3c

Fig. 3.1 shows the trend in ionisation energy for Period 3 of the Periodic Table.

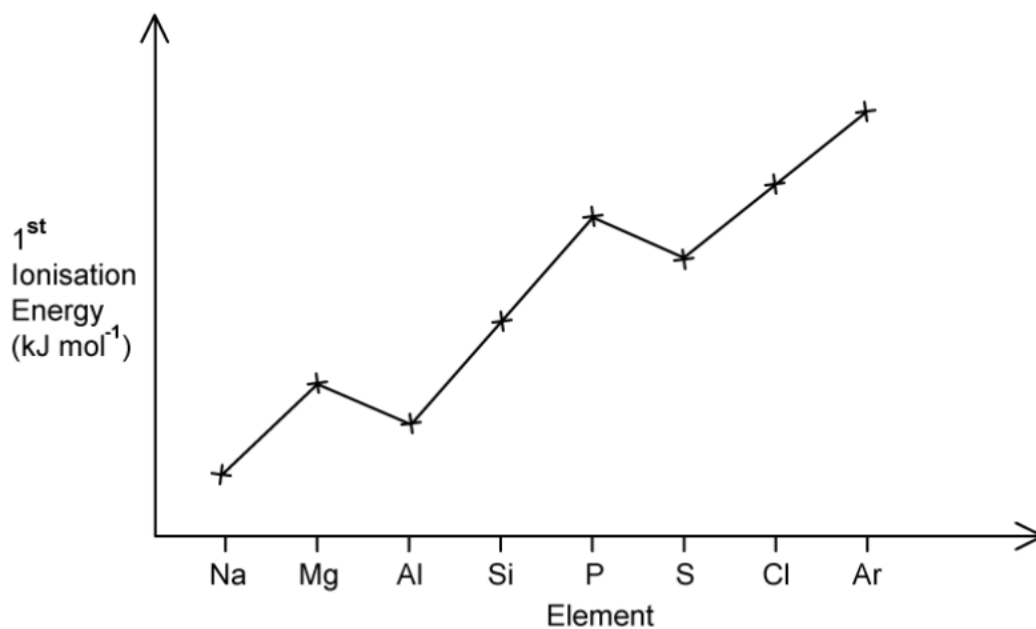


Fig. 3.1

i)
Explain why the first ionisation energy increases across Period 3.

[3]

ii)
Explain the deviations in the trend.

[3]

[6 marks]

Question 3d

Explain why the second ionisation energy of aluminium is a larger value than the first ionisation energy.

[1 mark]

Question 4a

Iron and cobalt are adjacent elements in the Periodic Table. Iron has three main naturally occurring isotopes, cobalt has one.

Explain the meaning of the term isotope.

[2 marks]

Question 4b

Isotopes of polonium, proton number 84, are produced by the radioactive decay of several elements including thorium, proton number 90. The isotope ^{213}Po is produced from the thorium isotope ^{232}Th .

Complete Table 4.1 below to show the atomic structures of the isotopes ^{213}Po and ^{232}Th .

Table 4.1

Isotope	Protons	Neutrons	Electrons
^{213}Po			
^{232}Th			

[3 marks]

Question 4c

Polonium has twenty five different isotopes.

Explain why the isotopes of polonium exhibit the same chemical reactions but their boiling points differ slightly.

[2 marks]

Question 5a

The first six ionisation energies of an element **X** are given below in Table 5.1.

Table 5.1

	First	Second	Third	Fourth	Fifth	Sixth
Ionisation energy (kJ mol ⁻¹)	950	1800	2700	4800	6000	12 300

Write an equation, with state symbols, for the second ionisation energy of element **X**.

[2 marks]

Question 5b

Use the data given above to deduce in which group of the Periodic Table element **X** is placed. Explain your answer.

Group

Explanation

[3 marks]

Question 5c

The first ionisation energies (I.E.) for the elements of Group 4 are given below in Table 5.2.

Table 5.2

Element	C	Si	Ge	Sn	Pb
1st I.E (kJ mol ⁻¹)	1090	786	762	707	716

Explain the trend shown by these values in terms of the atomic structure of the elements.

[4 marks]

Question 5d

Write the **full** electronic configuration for germanium.

[1 mark]

Model Answers: Medium

1a

a) The completed table is:

Atomic number	Nucleon number	Number of electrons	Number of protons	Number of neutrons	Symbol
9	19	10	9	10	${}^{19}_{9}\text{F}^{-}$
26	58	23	26	32	${}^{58}_{26}\text{Fe}^{3+}$

- Each correct row scores [1 mark]

[Total: 2 marks]

- You should know that:
- Atomic number
 - Number of protons in an atom
- Nucleon number
 - Sum of protons and neutrons
- Number of protons
 - Atomic number of the atom
- Number of neutrons
 - Nucleon number – atomic number (or number of protons)
- Symbol
 - In this question the nucleon number is on top and the atomic number on the bottom
 - Charge given as superscript
- If there are the same number of electrons and protons then the charge is neutral
- If the charge is positive then the atom has more protons than electrons
 - If the atom has lost two electrons then the charge is 2+
- If the charge is negative then the atom has more electrons than protons
 - If the atom has gained two electrons then the charge is 2–

1b

b) The symbol and full electronic configuration for the potassium +1 ion are:

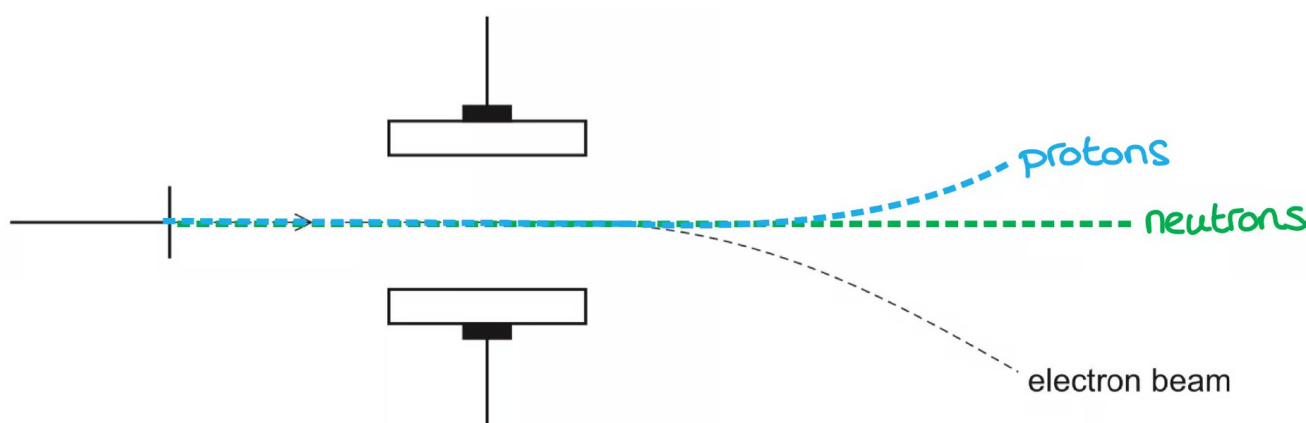
- Symbol: ${}^{39}_{19}\text{K}^{+}$; [1 mark]
- Full electronic configuration = $1s^2 2s^2 2p^6 3s^2 3p^6$; [1 mark]

[Total: 2 marks]

- The symbol should include the nucleon and atomic number
 - It does not matter which way around they are
- The full electronic configuration for the potassium atom is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
- One electron has been removed from the 4s shell to form the ion, so the electronic configuration will now be:
 - $1s^2 2s^2 2p^6 3s^2 3p^6$

1c

c) The diagram should include:



- Line straight on labelled 'neutrons'; [1 mark]
- Line (curving) up labelled 'protons'; [1 mark]
- Proton line clearly shows less (overall) deflection than electron curve; [1 mark]

[Total: 3 marks]

- When a beam of electrons is fired past the electrically charged plates, the electrons are deflected very easily away from the negative plate towards the positive plate – they have a small mass so are deflected very easily
 - The bottom plate is therefore positive and the top plate is therefore negative
- A beam of protons is deflected away from the positive plate and towards the

negative plate

- As protons are deflected less than electrons, this also shows that protons are heavier than electrons
- A beam of neutrons is not deflected at all which is why the line continues straight on and does not curve
- If you are asked to draw this in an exam, it may help to label the positive and negative plates so you do not get confused
- You also need to say protons and neutrons as more than one is being fired

1d

d)

i) The group number of element **Y** is:

- Group 16 / 6 / VI

AND

Big increase / big difference / big gap / big jump / jump in increase / jump in difference after 6th IE; [1 mark]

ii) The general trend in first ionisation energies across the third period is:

- Increases (across period) due to increasing attraction (of nucleus for electrons) due to increasing nuclear charge / atomic / proton number; [1 mark]
- Constant / similar shielding / same (outer / number of) shell / energy level; [1 mark]

iii) The first ionisation energy of element **Y** is less than that of element **X** because:

- Electron (pair) repulsion; [1 mark]
- (**Y** has a) pair of electrons in a (3)p orbital / a (3)p orbital is full; [1 mark]

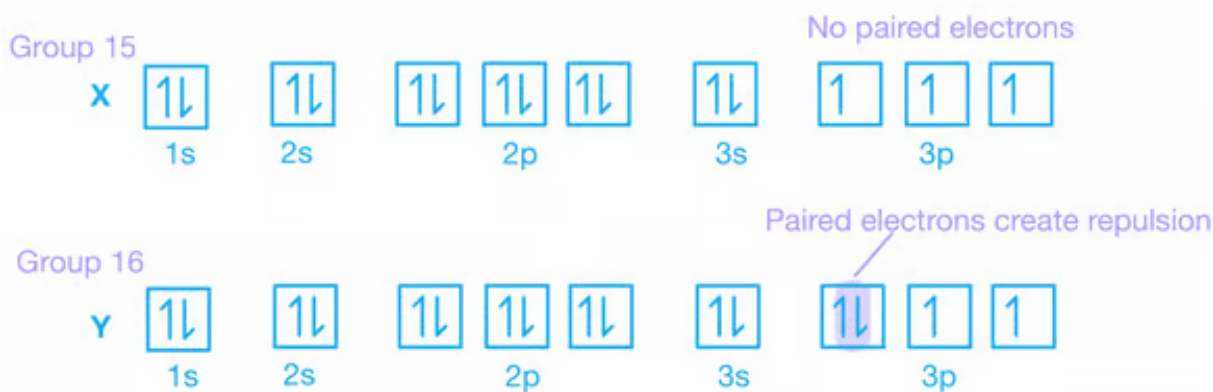
iv) The electronic configuration of element **Z** is:

- $(1s^2) 2s^2 2p^6 3s^2 3p^5$; [1 mark]

[Total: 6 marks]

- To determine the group number of an element from successive ionisation energy data you need to look for the large jumps
 - For element **Y** there is a large jump in ionisation energy between the 6th and 7th ionisation energies

- You should know why the trend in ionisation energy increases across a period and down a group
- The first ionisation energy of **Y** is less than **X** due to the pairing of electrons in the 3p orbital
 - You would expect the ionisation energies to increase across the period
- We can show this by looking at the electronic configurations:
 - X**: $(1s^2) 2s^2 2p^6 3s^2 3p^3$
 - X** is in Group 15 / 5 / V as there is a large jump between the fifth and sixth ionisation energies
 - Y**: $(1s^2) 2s^2 2p^6 3s^2 3p^4$
 - We know **Y** is in Group 16 / 6 / VI
 - $3p^4$ means the first electron will be removed from a 3p sub shell which contains **paired electrons**– they repel each other



- To determine the electronic configuration of element **Z**
 - You know that the element is in period 3
 - You can work out the group number by looking at the jump the ionisation energy
 - This is between the seventh and eighth ionisation energies, so element **Z** is in Group 17
 - The element in Period 3 and Group 17 is chlorine so the electronic configuration must be given for this element

2a

a) The completed table is:

Particle	Relative mass	Relative charge	Location within atom	Total number in
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				an atom of ^{197}Au
Electron	0.0005	-1	shell(s); [1 mark]	79
Neutron	1.001; [1 mark]	0; [1 mark]	nucleus	118; [1 mark]

[Total: 4 marks]

- This is a common question so make sure you know the correct information
- Note that protons are not given in this table
- **Remember:** To calculate neutrons: Mass number - atomic number so $197 - 79 = 118$

2b

b)

i) The term isotopes means:

- (Atoms of the same element) with the same proton / atomic number; [1 mark]
- (But) different number of neutrons / mass number; [1 mark]

ii) This different sample of gold has the same chemical properties as the sample found in the earth because:

- Same number of electrons / electronic structure; [1 mark]

[Total: 3 marks]

- It is a common question to be asked the definition of an isotope and why chemical properties are similar
- However, make sure you also know about why the physical properties may be slightly different
 - The only difference between isotopes is the number of neutrons
 - Since these are neutral subatomic particles, they only add mass to the atom
 - As a result of this, isotopes have different physical properties such as small differences in their mass and density

2c

c) The completed box notation is:

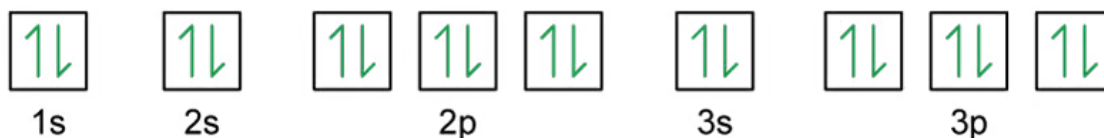
- Correct electronic configuration / all boxes contain two vertical arrows pointing in opposite directions; [1 mark]



[Total: 1 mark]

- Each box can only contain up to two single barbed arrows
 - If you use double barbed arrows you will also gain the mark
- Each of the arrows in the box must point in the opposite direction as this shows the opposite spin
- Make sure your arrows are vertical
- You have been asked for the chloride ion
 - The chlorine atom contains 17 electrons, the chloride ion contains 18

- Correct electronic configuration / all boxes contain two vertical arrows pointing in opposite directions; [1 mark]

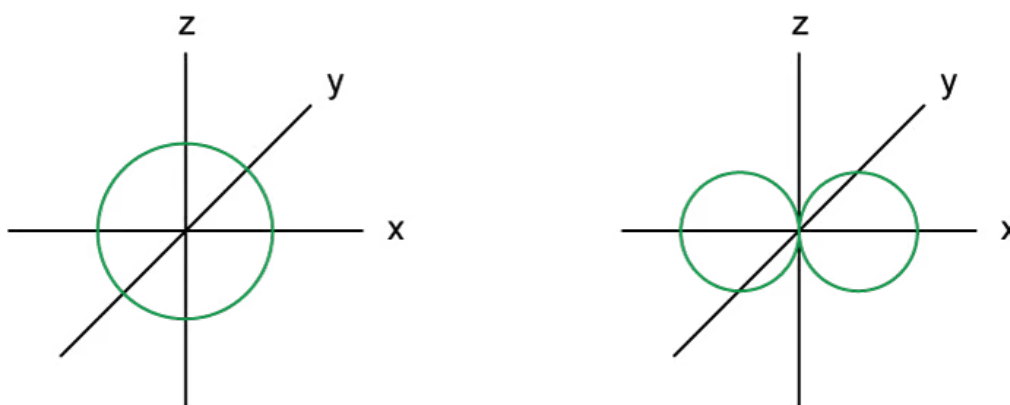


[Total: 1 mark]

- Each box can only contain up to two single barbed arrows
 - If you use double barbed arrows you will also gain the mark
- Each of the arrows in the box must point in the opposite direction as this shows the opposite spin
- Make sure your arrows are vertical
- You have been asked for the chloride ion
 - The chlorine atom contains 17 electrons, the chloride ion contains 18

2d

d) The shape of the orbitals in Period 3 are:



- Spherical s orbital; [1 mark]
- Double lobed p orbital drawn along any axis; [1 mark]
- Both labelled; [1 mark]

[Total: 3 marks]

- The s orbitals are spherical shaped and the p orbitals are dumbbell shaped
 - The electrons in the third period are occupied in these s and p orbitals
- You can draw the p orbitals along any of the axes as they can be p_x , p_y or p_z

3a

i) The definition of first ionisation energy of an element is:

- Enthalpy/energy (needed) to remove one mole of electrons from one mole of gaseous atoms; [1 mark]
- To form one mole of gaseous $1+$ ions / ions with $+1$ charge; [1 mark]

ii) The equation for the first ionisation energy of aluminium is:

- $\text{Al (g)} \rightarrow \text{Al}^+ \text{ (g)} + \text{e}^-$; [1 mark]
*State symbols **must** be included for the mark*

[Total: 3 marks]

- **Be careful** when writing the equation for ionisation energies
 - This equation is for the **1st** ionisation energy, because you are removing the **1st** mole of electrons and forming a mole of $1+$ ions
- A common mistake students make is to start with a $1+$ ion when it is the 1st ionisation energy, but this would be incorrect
 - The number of the ionisation energy is the order of the moles of electrons you are removing
 - 1st ionisation energy = 1st mole of electrons removed, forming $1+$ ions
 - 2nd ionisation energy = 2nd mole of electrons removed, forming $2+$ ions... and so on
- Another common mistake is to forget the state symbols – you **always** need to include state symbols for ionisation energy equations – you will lose the mark if you don't

3b

b) Element **A** is:

- Phosphorus / P; [1 mark]

- There is a (very) large jump between ionisation energies 5 and 6
OR
(Very) large jump in ionisation energy after the 5th ionisation energy; [1 mark]

[Total: 2 marks]

- Successive ionisation energies increase because once you have removed an electron from an atom, you are left with a positive ion
 - Trying to remove an electron from a 1+ ion is more difficult and requires more energy, removing an electron from a 2+ ion is even more difficult and requires even more energy... And so on
- The trick to determining an element from its ionisation energies is to look for the **really significant jump** in ionisation energy (IE)
 - For example, in this question the differences up to the 5th ionisation energy are not that significant, each IE increases by a similar amount
 - However, the jump from the 5th IE to the 6th IE is almost 15,000 kJ mol⁻¹ - very significant and much bigger than the other jumps up to this IE
 - This shows that while the 1st to the 5th ionisation energies will occur, the amount of energy required to remove another mole of electrons for the 6th ionisation energy is **too great and would not occur**
 - So, the element must be phosphorus
 - The electronic configuration is
 - 1s² 2s² 2p⁶ 3s² 3p³
- **Be careful:** a common mistake is to correctly spot that the 'big jump' is between the 5th and 6th ionisation energies, but to then incorrectly identify the element as being in Group 6
 - If the big jump happens after the 5th ionisation energy, that means that the 6th ionisation energy is too big, so the element is in Group 15 / 5 / V

3c

c)

i) The trend that this graph shows and the reasons for it are:

- A general increase in first ionisation energy (across Period 3); [1 mark]
- (Across a period) nuclear charge increases / a proton is added to the nucleus
AND
Shielding stays the same / shielding does not change / electrons are added to the

same shell; [1 mark]

- Atomic radius decreases

OR

Outermost / outer / highest energy / valence electrons are closer to the nucleus; [1 mark]

ii) The deviations from the trend are due to:

- Aluminium deviates because its outermost / highest energy / valence electron is in a 3p orbital; [1 mark]
- The 3p orbital is further away from the nucleus

OR

3p orbital has a (slightly) higher energy than an 3s orbital; [1 mark]

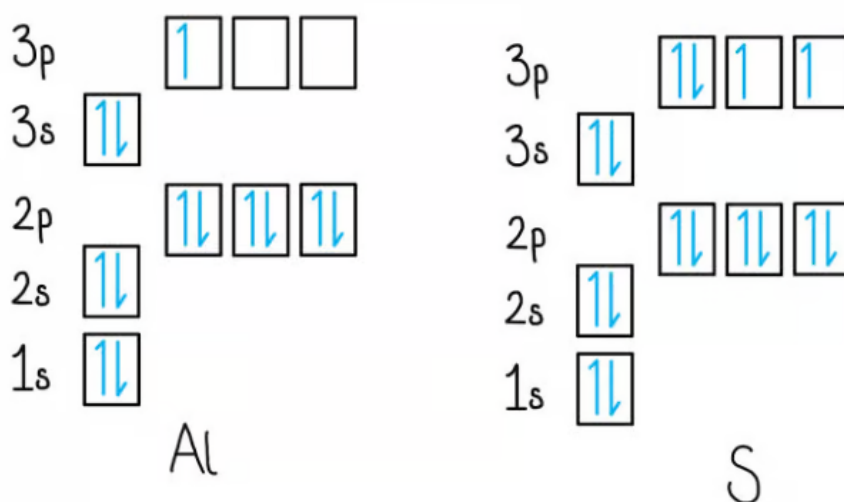
- Sulfur deviates because its outermost / outer / highest energy / valence electron is paired / shares an orbital with another electron / it is a paired electron which is being removed

AND

So there is repulsion between the electrons (in the orbital); [1 mark]

[Total: 6 marks]

- This is a common question to explain why Al and S deviate from the trend
- Make sure you state which orbital the outer electrons are in (i.e. 3s and 3p orbitals)



- The same marking points will also apply to Period 2 elements
 - The first ionisation energy increases across the period – for the same reasons
 - Boron has a lower first ionisation energy than beryllium and oxygen has a lower

first ionisation energy than nitrogen

3d

The second ionisation energy of aluminium is larger than the first ionisation energy because:

- Electrons are now being removed from positive ions

OR

The electrons being removed are closer to the nucleus

OR

Al⁺ ions are smaller than Al atoms

OR

Electron is being removed from a (3)s orbital rather than a (3)p orbital; [1 mark]

[Total: 1 mark]

- Successive ionisation energies increase in value because the more positive an ion is, the more energy is needed to remove electrons
 - You are now trying to remove a negatively charged electron from a positively charged ion
 - In the case of aluminium, you have the added effect that the 2nd electron is being removed from an s orbital rather than a p - this takes more energy
 - It is easier to remove electrons from a p orbital (on the same energy level) than it is to remove an electron from a s orbital
- As you remove electrons, the outer electrons are more strongly attracted to the nucleus and become closer to the nucleus
 - The stronger the attraction, the more energy is needed to remove electrons
 - So the higher the ionisation energy becomes

4a

a) The term isotope means:

- Same proton no. / atomic no. / no. of protons; [1 mark]
- Different mass no. / nucleon no. / no. of neutrons; [1 mark]

[Total: 2 marks]

- This is a definition that you should know

4b

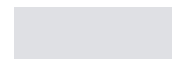
b) The completed table is:

Isotope	Protons	Neutrons	Electrons
^{213}Po	84	129	84
^{232}Th	90	142	90

- Each correct column scores [1 mark]

[Total: 3 marks]

- You are given the proton number for each isotope in the question
 - The number of protons in an atom is the same as the atomic number
- The number of neutrons = nucleon number - atomic number
- The number of electrons is the same as the atomic number unless there is a charge



^{232}Th	90	142	90
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- Each correct column scores [1 mark]

[Total: 3 marks]

- You are given the proton number for each isotope in the question
 - The number of protons in an atom is the same as the atomic number
- The number of neutrons = nucleon number - atomic number
- The number of electrons is the same as the atomic number unless there is a charge

4c

c) Isotopes of polonium exhibit the same chemical reactions but their boiling points differ slightly because:

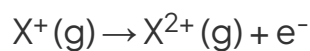
- Same number of electrons / electronic structure; [1 mark]
- Mass number differs / number of neutrons different; [1 mark]

[Total: 2 marks]

- Boiling point is an example of a physical property and isotopes of the same element will have slightly different physical properties but the same chemical properties
- Neutrons are neutral subatomic particles so they only add mass to the atom
- As a result of this, isotopes have different physical properties such as small differences in their mass, density, melting point and boiling point

5a

a) The equation for the second ionisation energy is for element **X** is:



- Correct equation; [1 mark]
- State symbols; [1 mark]

[Total: 2 marks]

- The second ionisation energy will form 2+ ions from 1+ ions
- Make sure you include the correct state symbols for the equation as this is a

fundamental part of the definition for ionisation energy

5b

b) Element **X** is in:

- Group 5 / 15; [1 mark]
- (Because there is a) sharp rise in successive ionisation energies between 5th and 6th IEs; [1 mark]
- Indicating change to a different shell / energy level

OR

Outer shell contains 5 electrons; [1 mark]

[Total: 3 marks]

- To determine the electronic configuration of element **X**
 - You know that the element is in Group 5 or 15
 - You can work out the group number by looking at the jump in the ionisation energy
 - This is between the fifth and sixth ionisation energies, so element **X** is in Group 5 or 15

5c

c) The trend is due to:

- Down the group atomic radii increase / outer electrons are increasingly further away; [1 mark]
- Electrons are added to new shells / more shells; [1 mark]
- More shielding; [1 mark]
- Despite increase in nuclear charge; [1 mark]

[Total: 4 marks]

- You should be able to explain why the ionisation energy decreases down a group
 - Make sure you break up your answer into clear marking points
 - You must say **nuclear** charge, not nucleus charge
- Overall
 - The number of protons in the atom is increased, so the nuclear charge increases

- But, the atomic radius of the atoms increases as you are adding more shells of electrons, making the atoms bigger

5d

d) The full electronic configuration of germanium is:

- $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^2$; [1 mark]

[Total: 1 mark]

- The question asks for the full electronic configuration so you must include all electrons
- You can place the $3d^{10} 4s^2$ either way round

